VPL: A Novel Cost Model for Earth and Mars-Orbit Navigation and Communication Constellations

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The problem

Limited public unit-cost data for navigation and communication constellations. Legacy models (SMAD USCM8, QuickCost) rest on data up to the early 2000s. We build a cost model on **25 systems** (1997-2025), that also cover the New-Space era.

The model

$$\mathrm{TOTAL}(M,N) = F\,g\,\Big[\underbrace{5.99\,M^{0.613}}_{\mathrm{DEV}(M)} + \underbrace{0.219\,M^{0.828}}_{\mathrm{PFM}(M)}\,N^{\mathrm{LRE}}\Big]$$

M = platform mass [kg], N = units, cost in FY2025 €M.

$\mathrm{LRE} = 1 + \frac{\ln 0.9}{\ln 2} = 0.848$. $g = 1$ traditional, 0.120 New-Space. $F = 1$ Earth, 1.14 Mars.

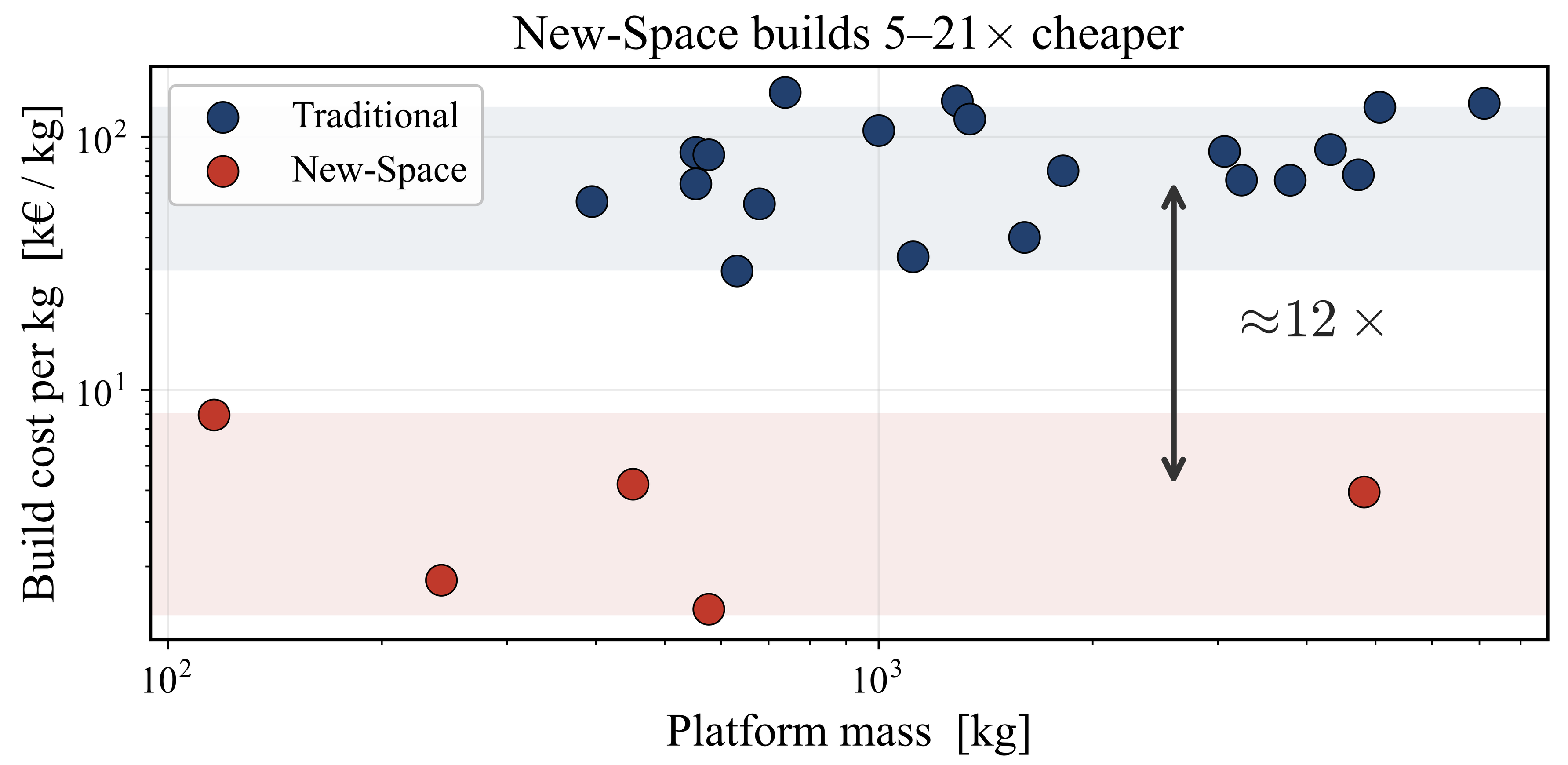
The main results

0.94 cross-validated R^2

5–21× New-Space cheaper

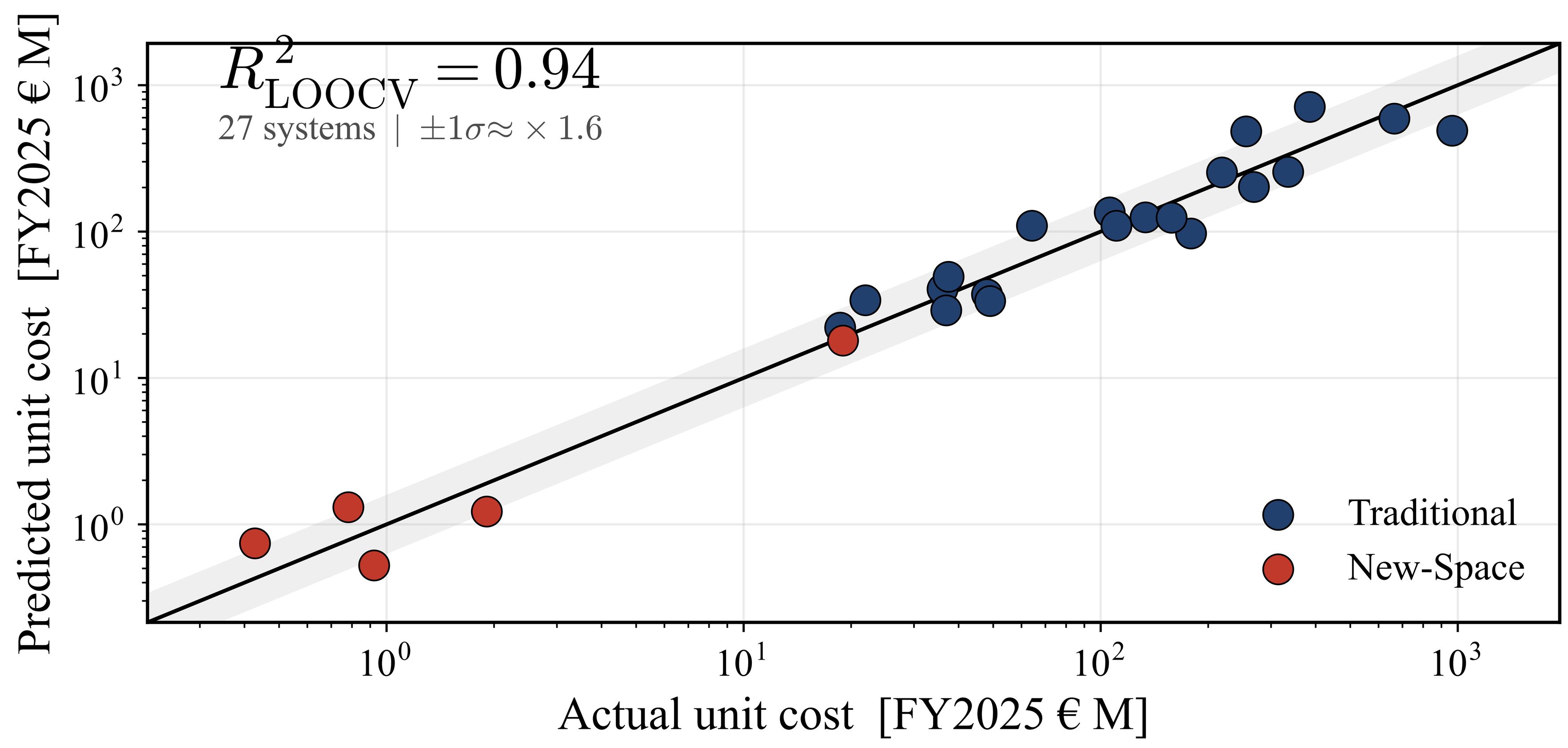
≈ USCM8+QuickCost on traditional; but SMAD over-predicts New-Space ~7×

Cost-plus vs new-space economy



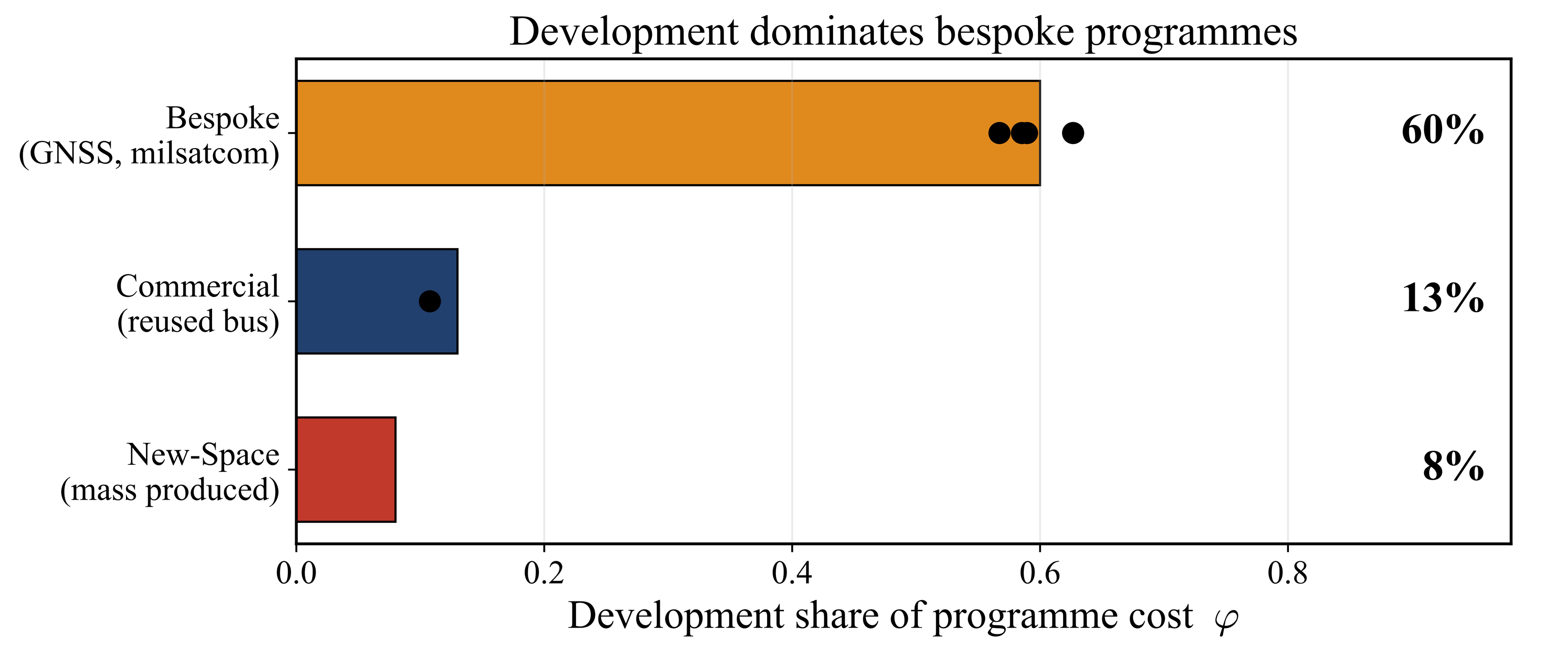
New-Space mass-produced builds are **5–21×** cheaper per kg, compared to traditional cost-plus bespoke systems.

CER model validation



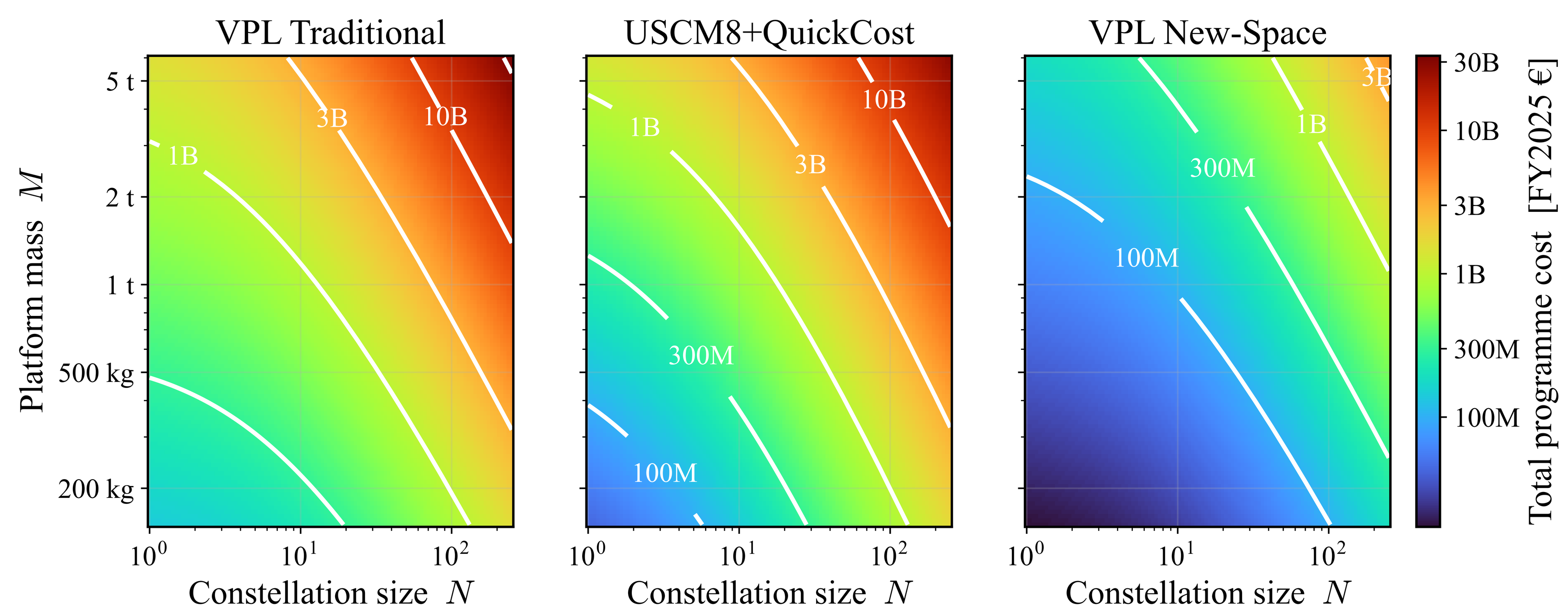
Leave-one-out cross-validation over 25 systems. The CER tracks unit cost across three orders of magnitude ($R^2 = 0.94$ pooled, 0.85 traditional-only).

Development vs production costs



The DEV/PFM split is fixed by disclosed programme budgets (US SARs, ESA).

Cost vs mass and constellation size



Iso-cost contours of $\mathrm{TOTAL}(M,N)$. VPL Traditional tracks the SMAD benchmark, with VPL New-Space a full band below.

Planetary cost factor

A subsystem bottom-up estimate gives a Mars bus-cost increase of **10% ± 4.5%**, adopted as $\mathbf{F = 1.14}$ (Earth $F = 1$). This replaces SMAD/QuickCost's blanket 1.29 planetary factor. Sources: RAND TR-418; Jones (ICES 2015).

Operations cost

Mission operations use the SME-SMAD Table 11-28 model: **€159 M** for 31 satellites over 5 years. Drivers: flight and ground software maintenance (SLOC), 19 engineers + 4 technicians, facilities and ground-hardware maintenance.

Data and code

Github repository with dataset, sources, python model, reproduction notebook, poster and a detailed report.

`github.com/cholevasm/vpl-cost-model`